

Chapter - 4

Java Conditionals

Sometimes we want to watch comedy videos on youtube if the day is sunday.

Sometimes we ordered junk food if it is our friend's birthday in the hostel.

you might want to buy an Umbrella if it's raining and you have the money.

you order the meal if also or your favourite drink is listed on the menu.

All these are decisions which depends on a certain condition being met.

In Java, we can execute instructions on a condition being met.

★ Decision making Instructions in Java

- if - else statement
- Switch statement

Examples

```
public class CONDITIONAL
```

```
{
    public static void main (String [] args)
```

```
{
    int age = 20;
    if (age > 18)
```

```
{
        System.out.println ("Yes, you can drive the car");
    }
```

```
    else
    {
```

```
        System.out.println ("No, you cannot drive the car");
    }
```

```
}
}
```

if-else Statement

The Syntax of an if-else Statement in C looks like that of C++ and Javascript. Java has a similar Syntax too it looks like:

```
if (Condition-to-be-checked)
```

```
{
```

```
    Statements - if-condition-true;
```

```
}
```

```
else {
```

```
    Statements if-condition-false;
```

```
}
```


Code Example

```
int a = 29;  
if (a > 18) {
```

```
    System.out.println("you can drive the car");  
}
```

NOTE: else block is optional if we don't use else also we can execute if statement.

• Relational Operators in Java

Relational operators are used to Evaluate Conditions (true or false) inside the if statements.

Some examples of relational operators are:

$=$, $>$, $<$, $<=$, $!=$
 $\downarrow$ $\downarrow$ $\downarrow$
 equals greater than equal to Not Equal

NOTE: '=' is used for assigned whereas '==' is used for equality checks.

★ Logical Operators

10 &&, ||, and ! are most commonly used logical operators in Java.

11 These are read as :

12

&&	→ AND	} Used to provide logic to our Java Programs
	→ OR	
!	→ NOT	

2 • AND OPERATORS

3 Evaluate to true if both the conditions are true.

4

Y && Y = Y	Y = True
Y && N = N	N = False
N && Y = N	
N && N = N	

```

6 public class LogicalOperators {
    public static void main (String [] args) {
        boolean a = true;
        boolean b = false;
        if (a && b) {
            System.out.println(a);
        }
        else {
            System.out.println(b);
        }
    }
}

```


• OR OPERATOR

Evaluate to true when at least one of the conditions a true

$$Y \parallel Y = Y$$

$Y = \text{true}$

$$Y \parallel N = Y$$

$N \rightarrow \text{false}$

$$N \parallel Y = Y$$

$$N \parallel N = N$$

```
public class LogicalOperators
```

```
{
    public static void main (String [] args)
```

```
{
    boolean a = true;
```

```
    boolean b = false;
```

```
    if (a || b)
```

```
{
```

```
        System.out.println (a);
```

```
}
```

```
    else
```

```
{
```

```
        System.out.println (b);
```

```
}
```

```
}
```

```
}
```


• NOT Operators

- 10 Negates the given logic (true becomes false and false becomes true.)

```
Public class LogicalOperators
```

```
{
```

```
    public static void main (String [] args)
```

```
    {
```

```
        boolean a = true;
```

```
        boolean b = false;
```

```
        System.out.println ("Not (a) is ");
```

```
        System.out.println (!a);
```

```
        System.out.println ("Not (b) is ");
```

```
        System.out.println (!b);
```

```
    }
```

```
}
```


★ Else-if clause

Instead of using multiple if statements we can also use else-if along with if thus forming an if-else-if-else ladder.

Using such kind of logic reduces indents last else is executed only if all the conditions fail.

```

1 public class Else-if
2 {
3     public static void main (String [] args)
4     {
5         System.out.println ("Taken Input From The User: \n");
6         Scanner sc = new Scanner (System.in);
7         System.out.println ("Enter the age value of a person: \n");
8         int age = sc.nextInt ();
9         if (age > 70)
10        {
11            System.out.println ("You are a Experienced Guy");
12        }
13        else if (age > 50)
14        {
15            System.out.println ("You are a Semi-Experienced Guy");
16        }
17        else if (age > 30)
18        {
19            System.out.println ("You are a Semi-Semi-Experienced Guy");
20        }
21    }
22 }
  
```



```

else
{

```

```

    System.out.println("you are not Experiencing");
}
}
}

```

● Switch Case Statements

Switch case is used when we have to make a choice b/w number of alternatives for a given variable.

```

public class SWITCH_CASE
{

```

```

    public static void main(String[] args)
    {

```

```

        System.out.println("Taken Input From The Users:");

```

```

        Scanner sc = new Scanner(System.in);

```

```

        System.out.println("Enter Your age:");

```

```

        int age = sc.nextInt();

```

```

        switch(age)
        {

```

```

            case 20:

```

```

                System.out.println("You are going to become adult.");

```

```

                break;

```

```

            case 40:

```

```

                System.out.println("You are going for job Experience.");

```

```

                break;

```


case 60:

System.out.println("you are going to
retired from job");

break;

case 60:

System.out.println("you are going to
take rest for your life");

break;

default:

System.out.println("All the best in
your life");